ENGR 111
Introduction to Engineering
Class #7

How people learn

How learning works, so you can get better at it

Zoom protocol

Name Set your Zoom name to your real name, or whatever you'd like me to call you.

Photo Set a professional profile photo.

Camera Encouraged, not required.

Mic Off unless you're speaking.

Your questions, anytime Type in chat or raise your hand, and I'll check periodically. Or just unmute and jump in.

My questions to the class Answer in chat, or unmute and answer.

If you speak Please state your name first.

If I drop out (e.g., my Wi-Fi fails) Stay in the meeting. I'll be back in a few minutes on my phone hotspot.

Where we are today

Questions we will explore here.

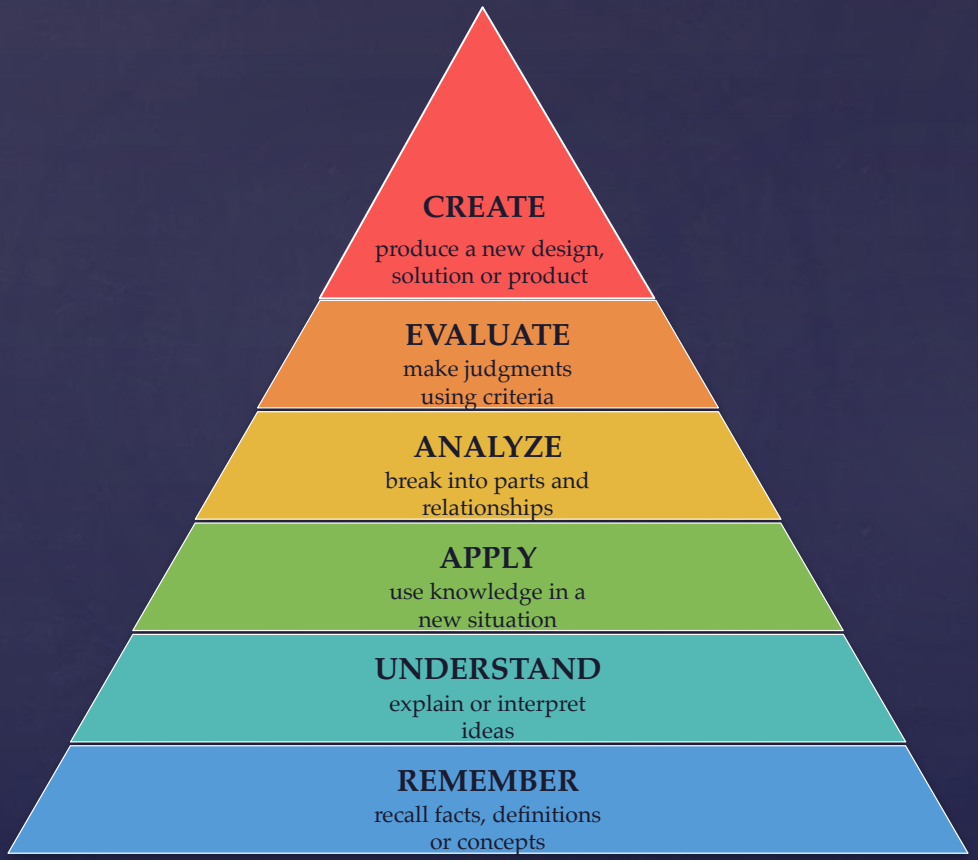
- How do we learn?
- How deeply are we engaging with and using what we learn?
- What makes learning stick after the lecture is over?

Why we are covering this?

- Engineering courses don't teach "how to learn", but it forms the basis of your entire engineering education.

Bloom's taxonomy

Framework for categorizing educational goals (A way to name how deeply you can use what you learned).



The idea. Learning can mean more than remembering a fact. It can mean explaining it, using it, taking it apart, judging it or making something new.

Why it matters. A course can ask for any level. The verb usually tells you which one: list, explain, solve, compare, defend, design.

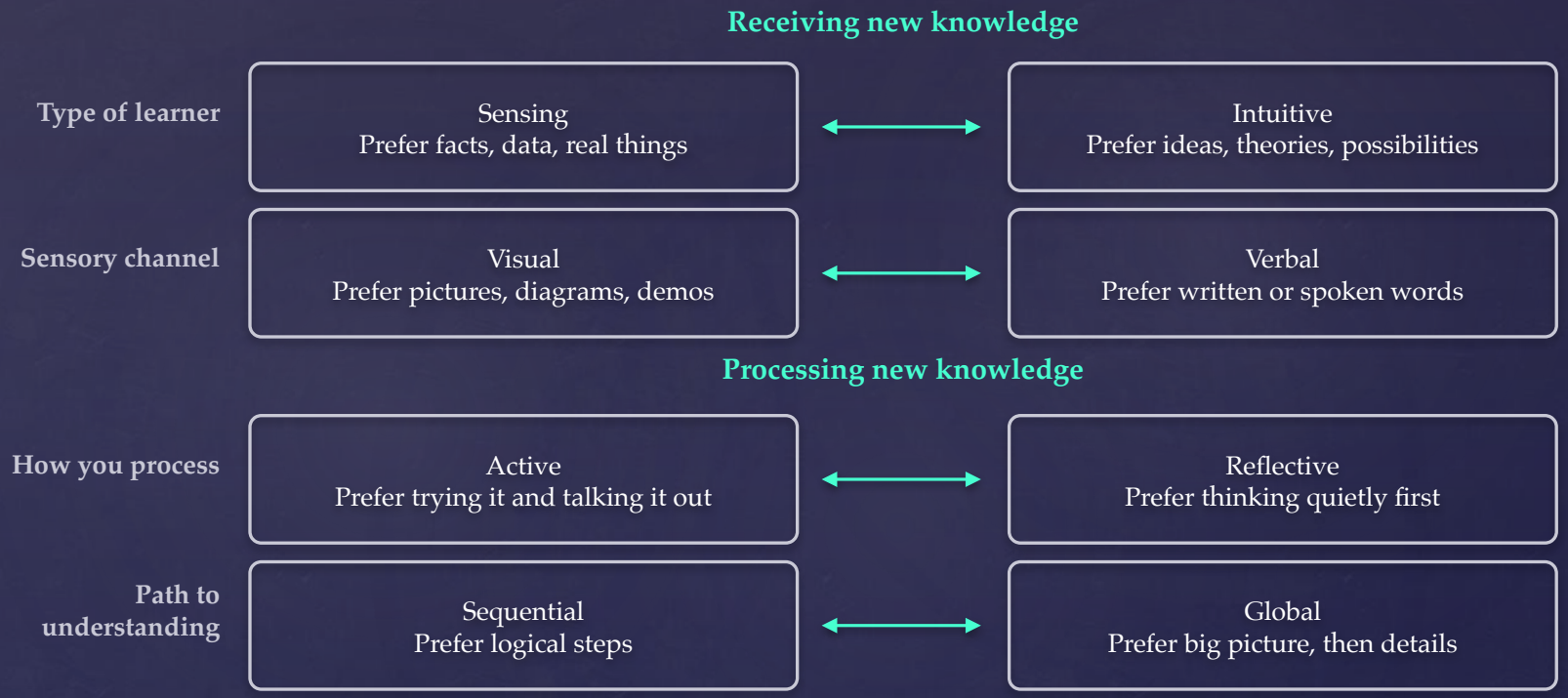
The version shown here. The 1956 taxonomy was revised in 2001. This is the common revised pyramid.

In college. What you do after class is not extra. It is most of the course and where most of the learning comes from.

Suppose you listen to a one hour podcast on the strangeness of quantum mechanics, what level on the pyramid is your learning you think?

Four learning-style scales

Learning styles: How do you prefer to receive and process new knowledge?



The key word is "prefer". The side that feels natural first, not the only one you can use.

How do you measure on all four scales?

Which side of each scale does a traditional engineering lecture feed?

Use the scales as a checklist

The Index of Learning Styles gives a score on each of the four scales.

https://learningstyles.webtools.ncsu.edu

The questionnaire. Forty-four choices. It reports whether you are strong, moderate or balanced on each scale.

Support your preferred learning styles. It can help you compensate when the class is taught in a way that does not fit you.

Use the opposite side too. It gives another perspective on the material and builds skills you would otherwise skip.

The textbook. All learners benefit from using both sides of all four dimensions.

Learning is reinforcement

The textbook's principle: repeated exposure over time is how material sticks.

Repeated exposure. You do not learn mechanics, calculus or chemistry in one pass. You meet the ideas again and again.

Each pass counts. Before class, in lecture, after class, before the exam.

Cramming. It can get you through an exam. It will not stick for the next course.



Spaced review changes the curve

Review soon, then again later, before the material has to be relearned.

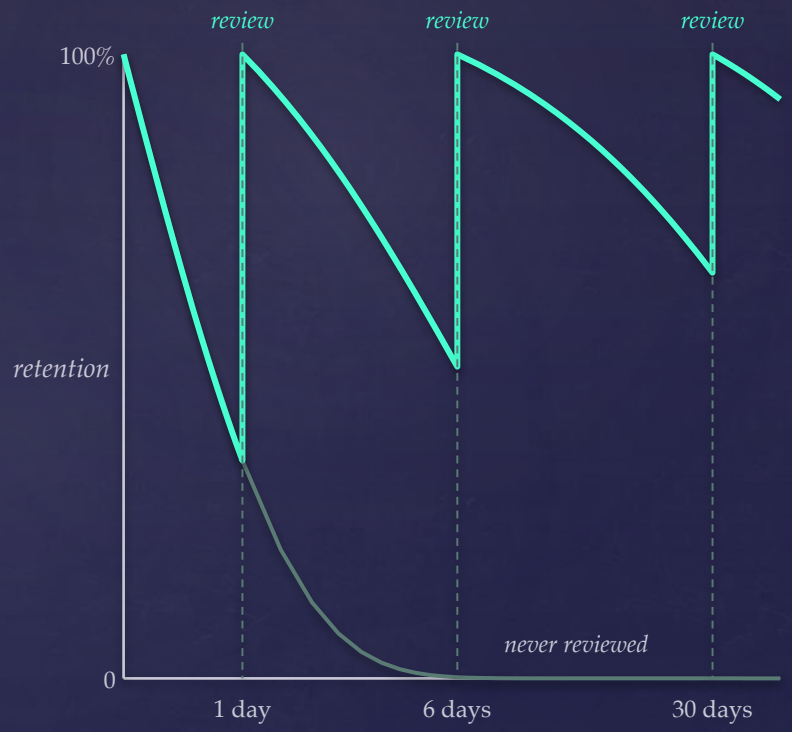
After one day. The textbook says you can forget over 60 percent of what you studied.

Through review. You can regain close to 100 percent, then retain the material longer.

The reviews get shorter. About ten minutes within a day, five minutes around six days, and two to four minutes around thirty days.

Use this to prepare for exams. Review in multiple short sessions before an exam (not just in one cram session the night before).

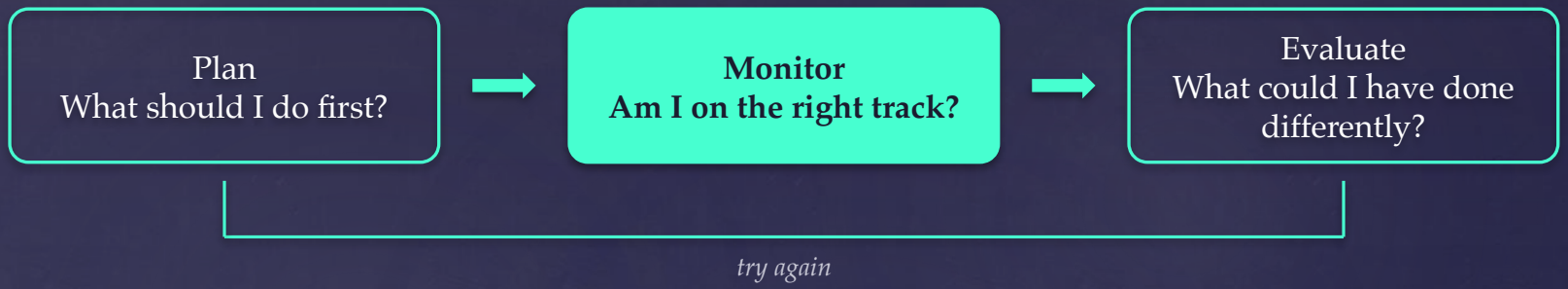
No reviews. Studying for the exam becomes relearning, and the next course gets harder.



Schedule the first review while the lecture is still fresh.

Metacognition

Watch your own learning: plan it, monitor it, evaluate it.



Observation	Change
I can't do problem 5.	Seek help: office hours or a classmate.
I keep rereading the same page.	Close the notes and test myself.
I'm lost in lecture.	Ask a question right then.

When you notice you are not learning, change the approach.

Before, during, and after a lecture

Preparing first makes the lecture your second pass, not your first.

BEFORE

Read the next section, even quickly.

Try two problems from the end of the chapter.

Skim the last set of notes.

Write down one question you want answered.

DURING

Sit where you actually pay attention.

Notes by hand. Slower means more selective, and it sticks better.

Ask when you are lost.

AFTER

Annotate your notes within a day.

Work the assigned problems, then one that was not.

Go over the problems with a study partner.

Explain one idea out loud to somebody else.

Pick one item from one column and do it this week.

What to take from today

Bloom. Learning is not only remembering. Courses ask you to apply, analyze, evaluate and create.

After class. Most college learning happens outside class.

Learning-style scales. Know your preferences, then study on both sides of each scale.

Reinforcement. Review in short, spaced passes instead of cramming.

Metacognition. Watch your own learning and change the method when it is not working.

Lectures. Prepare before, ask during, work problems after.

Pick one change small enough that you will actually do it.